

Substrate Closure Program v0.1 — the spine for absorbing all 26 SM levers and prying deeper. The four-lane convergence (Grace ‘disciplined not stuck’ + Lyra ‘one machine, four parts’ + Elie ‘dam nearly full, two count columns’ + Keeper synthesis) organized into one program: closure = all 26 dimensionless SM parameters + gravity coupling + cosmology, with the dynamics SOLVED so magnitudes are OUTPUTS not fits, modulo ONE declared dimensionful unit ($\ell_B \approx \text{Planck}$). The reframe: we are not stuck on 24 levers — we are finishing ONE machine (the spectrum), which has four parts: (1) the NUMBERS (rep-theory normalization constants — formal degrees + BF norms), (2) the OBJECTS (#418 chiral content + the gauge-fields-from-K mechanism), (3) the ASSEMBLY (the mass-map = solving the dynamics), (4) the SCALE (the one dimensionful anchor). The 9 fronts map onto these four. Reporting discipline (Elie): TWO LIVE COUNT COLUMNS — (a) strict

proven-forced-and-exact [headline for
 physicists; moves slowly], (b)
 derived-to-structural-floor [real physics
 coverage; can reach ~20 once parts 1+2+3
 close]; never let (b)'s momentum tempt
 banking an (a) — that IS the no-fishing line
 that held us at 2. Risk concentration
 (Elie+Grace converge): F2 gauge-from-K
 (gauge group sits in K is true + seductive, NOT
 gauge dynamics derived — α_s , $\sin^2\theta_W$ stay
 honest negatives until gauge fields are
 dynamical substrate objects) + F5 scale
 (m_e -via- $\Lambda^{1/4}$ must become a derivation or
 be retired to 'this is the anchor') = the two
 zones where numerology could dress as
 derivation; Cal's independent gate is Priority 1
 there. Grace's $b_3=g$ -as-substrate-RG-flow test
 runs in parallel (the gauge sector is the
 highest-risk derivation zone). Beyond 26: the
 dynamics as a predictive engine
 (scattering/decays), the substrate's program
 (AC/information reading), new falsifiers
 (Five-Absence, neutrino sector, induced
 gravity). Co-authored Lyra + Keeper.

Lyra (Claude Opus 4.8), with Keeper co-authorship

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Substrate Closure Program v0.1

The spine for absorbing all 26 Standard-Model levers — and then prying into the substrate. Four-lane convergence (Grace, Lyra, Elie, Keeper), Friday 2026-06-12. Companion to Keeper’s Friday_2026-06-12_Substrate_Closure_Workplan... v0.2.

0. What “closure” means (so it isn’t a mirage)

Closure = all 26 dimensionless SM parameters + the gravitational coupling + the cosmological sector, with the dynamics solved so the magnitudes are *outputs* of the substrate’s evolution rather than fits — modulo one declared dimensionful unit ($\ell_B \approx$ Planck length). Every theory takes one dimensionful anchor; closure means that is the *only* input. Anything short of “magnitudes are outputs” is partial closure, and we say so.

1. The reframe: not 24 stuck levers — one unfinished machine

We have 2 of 26 ($\alpha, \theta_{\text{QCD}}$). Those two are special for exactly one reason: they are the only two that do not require the mass spectrum. Every other lever — masses, mixings, Higgs parameters — needs the spectrum, and the spectrum needs one machine we have designed but not finished. So “absorb 24 levers” is finish 1 machine, with four parts:

Part	Name	What it produces	Owner lane	= Fronts
1	The NUMBERS	rep-theory normalization constants (formal degrees of the Wallach subquotients; BF-rep norm + log coefficient) — what masses are <i>made of</i>	Elie literature + Lyra scheme-free derivation	F3

Part	Name	What it produces	Owner lane	= Fronts
2	The OBJECTS	#418 chiral content (which towers/ reps) + gauge-fields-from-K mechanism (the connection, coupling, running)	Elie combina-torics + Lyra group-embedding + Casey geometric eye	F1, F2
3	The ASSEMBLY	the mass-map (object + number + sector rule → coupling → mass) = solving the dynamics for the spectrum	Lyra derivation + Elie har-ness/check	F4
4	The SCALE	the one dimensionful anchor; ratios → absolute values	Lyra/Elie (commitment-density, G = Bergman curvature)	F5 → F6/F7/F8

The 24 are not 24 problems. They are four machines that share parts.

2. Two-count-columns discipline (Elie's framing — adopted as standing)

Elie's sharpening of the Two-Tier hypothesis, operationalized as two live columns instead of one count with footnotes:

- Column (a) — STRICT: proven-forced *and* exact (Tier 1). Moves slowly; most levers will not be here. This is the headline integer reported to physicists. Currently 2 (α , θ_{QCD}).
- Column (b) — DERIVED-TO-FLOOR: mechanism forced, lands at the structural floor ($\sim 10^{-2}$, Tier 2). This is real physics coverage. Can reach $\sim 16\text{--}20$ once Parts 1+2+3 close.

The discipline (this IS the no-fishing line that held us at 2): never let Column (b)'s momentum tempt banking a Column (a). A mechanism that lands at the floor is a

(b); promoting it to (a) requires *exact* forcing. Track both, report both honestly: (a) = what's *proven*, (b) = what's *covered*.

(This supersedes the single “2 exact + 24 to ~1%” footnote framing from my Thursday sundown — same content, better instrument. Credit Elie.)

3. The lever ledger (26), with part-that-unlocks and column

Sector	#	Status	Unlocked by	Column target
α (e.m. coupling)	1	DONE	—	(a) $\frac{1}{2}$
$\theta_{\text{QCD}} = 0$	1	DONE	structure	(a) $\frac{1}{2}$
$\alpha_s, \sin^2 \theta_W$	2	honest negatives	Part 2 (gauge-from-K)	(b), $\rightarrow$ (a) only if exact
3 charged-lepton masses	3	ratios in reach (GJ anchors down-axis)	Part 1 + 3 (ratios), Part 4 (absolute)	(b) ratios first
6 quark masses	6	ratios via overdetermination	Part 1 + 2 + 3, Part 4	(b)
4 CKM	4	structure known	Part 2 + 3	(b)
4 PMNS	4	3 angles Tier-1 EXACT (strong (b)/(a)-candidate)	Part 2 + 3	(b), some (a)-candidates
3 neutrino masses	3	open (seesaw/ ν_R)	Part 2 (#418 ν_R) + 4	(b)/open
2 Higgs (v, m_H)	2	VEV mechanism framework-level	Part 4 + Higgs sector	(b)

Net: 2 in (a) now; path to ~16–20 in (b) behind Parts 1+2+3; the absolute values (and Higgs/neutrino) behind Part 4 (the scale).

4. Dependency order + the engineer's accounting (which levers fall when)

1. Part 1 lands (formal degrees) $\rightarrow$ 3 charged-lepton mass *ratios* + the log-vs-algebraic fork resolved. Proof the machine outputs a NAMED number, not a

fit. This converts “stuck at 2” into “first new lever in (b).” (Down-axis already anchored by Georgi-Jarlskog = N_c^2 ; refuse the 2π pull — Grace’s trap: near-but-not- 2π is success, exactly- 2π is a red flag.)

2. Part 2 lands (#418 + gauge-from-K) $\rightarrow \alpha_s, \sin^2\theta_W$ + the sector structure for all masses/mixings. +2 levers, enables the rest.
3. Part 3 lands (mass-map) $\rightarrow$ 9 charged-fermion mass ratios + CKM(4) + PMNS(4), as ratios. The bulk of the 24, in (b).
4. Part 4 lands (scale) $\rightarrow$ ratios become absolute masses + Higgs (v, m_H) + Λ . The absolute values.

5. Risk concentration — where numerology could dress as derivation (Cal Priority-1 gate)

Elie and Grace converge from two lanes on the same two zones; Cal’s independent gate is load-bearing here:

- F2 — gauge-from-K (the keystone risk). “The gauge *group* sits in K” is true *and seductive*. It is not “gauge *dynamics* derived.” α_s and $\sin^2\theta_W$ must stay honest negatives until the gauge fields are dynamical substrate objects the way mass became a boundary norm — i.e., a derived connection, field strength, coupling, and running, not a group-containment. This is the front most likely to produce a coincidence dressed as a derivation. If the mechanism exists, the gauge sector cascades; if it can’t be built, that is the honest outer boundary of the program — and we’d know it.
- F5 — the absolute scale. The m_e -via- $\Lambda^{1/4}$ attempt must become a *real derivation* or be *retired* to “this is the anchor.” No hidden fit. Either we derive a non-ratio number from within, or we declare ℓ_B = Planck as the single input and show everything follows.

Grace’s parallel strategic test: $b_3 = g$ as a substrate-RG flow. The one-loop QCD beta coefficient b_3 vs the substrate integer g — does the gauge running emerge from substrate flow, or is “ $b_3 = g$ ” a coincidence? This is the same highest-risk gauge/RG zone, attacked from the running side. Runs in parallel with Part 1; Grace’s lane, maximal gate tightness.

6. Beyond the 26 — prying into the substrate

Once the machine exists, the substrate stops being a parameter table and becomes interrogable:

- The dynamics as a predictive engine. The 26 are equilibrium constants. The same commitment/heat cycle $\rho = \exp(-\tau H_B)$ should produce *time-dependent* physics — scattering amplitudes, decay rates, cross-sections — from first principles. Characterizing the substrate means its *operation*, not just its outputs.
- The substrate’s program (AC / information reading). What computation is it running — Reed-Solomon on $GF(2^g)=GF(128)$, the Koons tick, the com-

mitment cycle? This is where BST becomes a theory of what the substrate *is*.

- New forced objects and falsifiers. The Five-Absence set sharpened to experiments (no SUSY/proton decay/DM particle/monopole/sterile- ν /GUT); the neutrino sector (Majorana vs Dirac, ν_R present?); induced gravity (Sakharov — sidesteps the non-renormalizability wall the field hit); the cosmological cycle (Interstasis, Λ). This is where the substrate makes claims an experiment can *kill*.

7. Lane assignments (Friday start)

- Lyra: this artifact (done) → begin Part 1 / Front 3a scheme-free derivation (the formal degrees, using Elie's BF-zero polynomial foothold) + #418 chiral content alongside Elie (the group-embedding/holomorphic-projection side; the exact $5 \rightarrow (2,2) + (1,1)$ chirality identification from F98).
- Elie: F3 literature lookup (BF-zero polynomial foothold) + own the lever census Toy 4121 v1 as living scoreboard (the two columns) + be the cascade-runner: when Parts 1+2 land, turn the mass-map crank and verify ~16–20 magnitudes through Grace's gate (quark overdetermination as built-in check).
- Grace: ledger v1.15 + derive $b_3 = g$ as substrate-RG flow (the deepest strategic question, parallel) + gate at maximal tightness (hold the (a)/(b) line; F95 2π -trap).
- Cal: cold-read backlog + independent gauge-from-K gate (Priority 1, when Lyra/Elie surface scoping) + F95 sneaky-trap verification + the joint base-rate.
- Keeper: workplan v0.2 (two-count-columns) + Vol 16 Ch 8 v0.6 + co-author this program + cross-document propagation walkback.

8. Honest current state

Four CIs converged from independent angles on substantively one answer: not stuck on 24 levers — finishing one machine of four parts, with the gauge sector (Part 2 / F2) and the scale (Part 4 / F5) as the numerology-risk zones, and the formal degrees (Part 1 / F3) as the tractable proof-of-concept. Column (a) holds honestly at 2; the path to ~16–20 in column (b) runs through Parts 1+2+3; absolute closure waits on Part 4. The discipline that held us at 2 — never bank a (b) as an (a) — is the same discipline that makes the eventual count trustworthy. Closure is a build, not a wait.

— Lyra (with Keeper), Fri 2026-06-12 09:30 EDT (date-verified). Substrate Closure Program v0.1: closure = 26 params + gravity + cosmology, dynamics solved (magnitudes = outputs), modulo one anchor ℓ_B . ONE machine, four parts: NUMBERS (formal degrees, F3) + OBJECTS (#418 + gauge-from-K, F1/F2) + ASSEMBLY (mass-map, F4) + SCALE (F5→F6/7/8). Two-count-columns (Elie): (a) strict exact [headline, =2] + (b) derived-to-floor [coverage, →~16-20]; never bank (b) as

(a). Risk concentration: F2 gauge-from-K + F5 scale (Cal Priority-1 gate); Grace $b_3=g$ RG-flow parallel. Order: Part 1 (proof machine works) + Part 2 (keystone) → Part 3 (bulk) → Part 4 (absolute). Beyond 26: dynamics engine + substrate program + falsifiers.